Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2024	ENGI3291-WE01

Title: (BEng/MEng)

Thermodynamics and Fluid Mechanics 3

Time Allowed:	2 hours
Additional Material provided:	No
Reporting problems during the exam:	Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk. If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk.

Instructions to Candidates:	This is a synchronous, time restricted exam. All submissions must be uploaded to Gradescope as per the published timetable. Answer ALL questions. Please answer each question on a new page. All workings must be shown. This is an online open-book exam. • *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the exam submission window.* • *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.*

Revision:	

Q1

(a) A repeating stage turbine is designed to have constant axial velocity and zero swirl exit velocity.

(i) Sketch a cascade view of the stage and the velocity triangles. In the cascade view, include the known velocity vectors and the numbering system you choose to use.

[20%]

(ii) Starting from the velocity triangles derive the relationship:

$$R = 1 - \frac{\psi}{2},$$

where R is the reaction, ψ is the stage loading coefficient.

[20%]

(iii) Show that for a *non-zero* swirl exit velocity, while other conditions remain unchanged, the reaction can be generalised to be:

$$R = 1 - \frac{\psi}{2} \quad \frac{\tan\alpha_2 + \tan\alpha_3}{\tan\alpha_2 - \tan\alpha_3}\Bigg),$$

where α_2 is the stator exit flow angle and α_3 is the rotor exit flow angle.

[30%]

(b) A steam turbine consists of 5 repeating stages with the turbine inlet stagnation temperature of 776 °C and stagnation pressure of 20 bar. It operates with a mass flow rate of 35 kgs^{-1} and generates 14 MW of mechanical power. The total to total efficiency of this turbine is 75%. The stator in each stage turns the flow from -15° at stator inlet to 70° at stator outlet. The turbine mean radius is 0.9 m and the rotational shaft speed is 3000 rpm. Assume constant axial velocity throughout the stages.

(i) Estimate the turbine outlet stagnation pressure.

[10%]

(ii) Determine the flow coefficient. Calculate the turbine stage loading coefficient and hence the reaction.

[20%]

continued

Q2

(a) Name three isotopes of the atom of Hydrogen. Are they stable or radioactive? If any isotope is radioactive, describe the type of radioactive decay it undergoes and provide the corresponding reaction(s).

[10%]

(b) Explain the difference between fissile, fissionable and fertile materials. Give an example in each case.

[10%]

(c) What are the most likely outcomes of neutron interaction with matter? Describe each outcome, provide its reaction equation and explain how the probability of each outcome can be quantified.

[15%]

(d) A nuclear reactor of size 250 cm $\times$ 100 cm $\times$ 100 cm operates at a power level of 25 MW. The average energy released per fission is 200 MeV and macroscopic cross-section for fission is $\Sigma_f = 0.05$ cm^{-1}. What are the fission reaction rate and the neutron flux in the reactor? The unit of energy on atomic scale is 1 eV $= 1.6 \times 10^{-19}$ J.

[15%]

(e) Describe the main characteristic features of the AGR reactor and explain why it has a large size compared to other technologies.

[15%]

(f) Sketch typical steam expansions through the turbines of 'Sub-Critical' Large fossil fired steam cycle on the T-s diagram provided (Fig. Q2), and using the data in Table Q2. Comment on the steam conditions at the inlets and at the exits from the steam turbines.

	Typical Sub-Critical Fossil Unit Steam Cycle Conditions	Typical PWR Nuclear Steam Cycle Conditions
HP inlet Pressure	160 bar	65 bar
HP inlet Temperature	540 °C	Steam is just wet, $< 1\%$, moisture content ($T_{sat} = 281$°C)
HP Exhaust Temperature/Condition	330 °C	Steam 13% Wet
HP Exhaust/Reheater Inlet Pressure	44 bar	11.7 bar
IP Inlet/Reheater Exit Pressure	40 bar	n.a.
IP Inlet Temperature after Reheat	540 °C	n.a.
LP inlet pressure	10 bar	11.4 bar
LP inlet Temperature	350 °C	270 °C
LP Exhaust Pressure	0.075 bar	0.05 bar
LP exhaust wetness/temperature	8%	10%
Steam Mass Flow	500 kg/s	1600 kg/s
Power output	600 MW	1000 MW

Table Q2: The operation conditions/parameters of a Sub-Critical Fossil Unit Steam Cycle, and an AGR (EdF) Nuclear Steam Cycle.

[20%]

continued

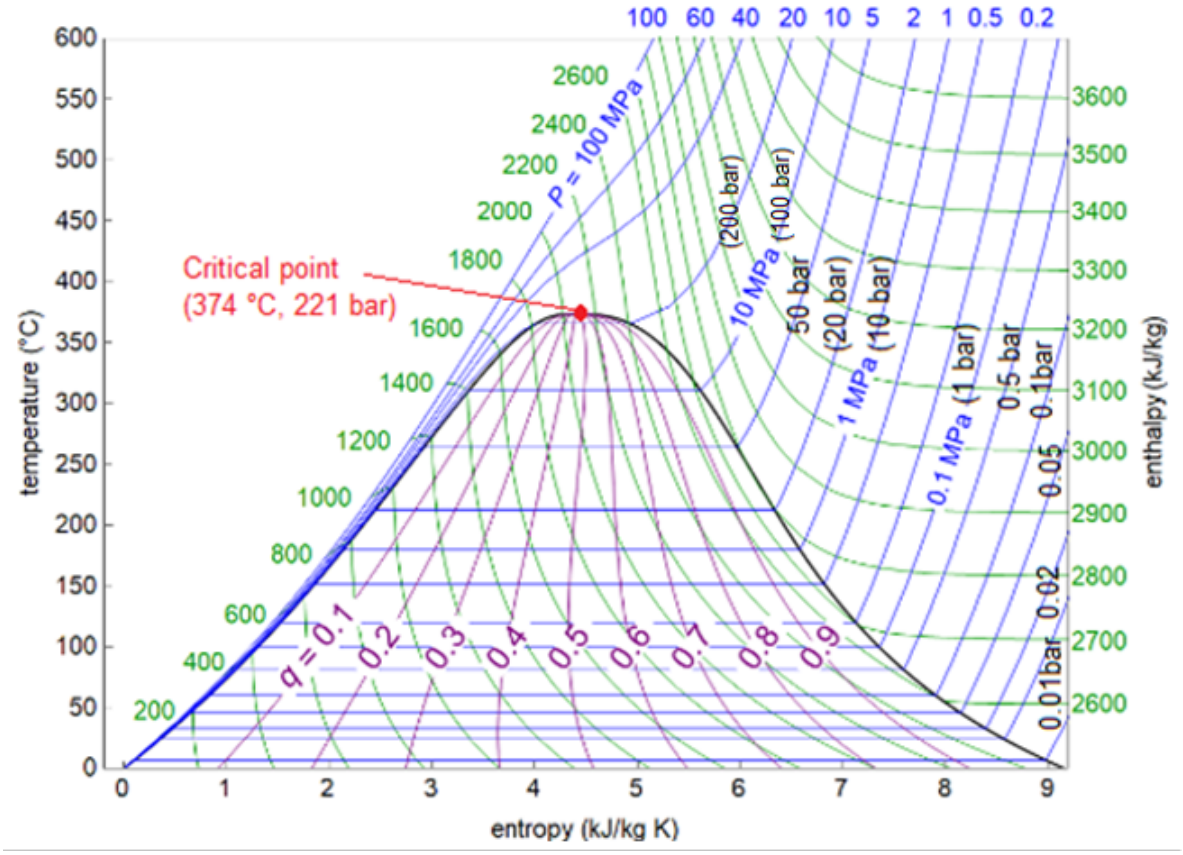


Fig. Q2

(g) Compare the typical electric power outputs from Sub-Critical Large fossil fired steam cycle and PWR nuclear power cycle. Explain the differences in the steam outputs from the in these two types of cycles, that produce the different power output levels.

[15%]

Q3

An aircraft exhaust system shown in Fig. Q3, consists of a duct of constant cross-sectional area A from 1-2 followed by a convergent-divergent nozzle 2-4. The area at the throat is A_T and the divergent nozzle exit area is A_D. Both the duct and nozzle are frictionless. The stagnation temperature and stagnation pressure at the inlet 1 are $T_{01} = 350$ K and $p_{01} = 12$ bar, respectively and the static pressure at ambient is $p_a = 1$ bar. The inlet Mach number is maintained at $M_1 = 0.15$. Further it is required that there are no shocks within the nozzle in any operation condition, i.e. the nozzle part, after 2, is isentropic. The flow can be treated as a perfect gas.

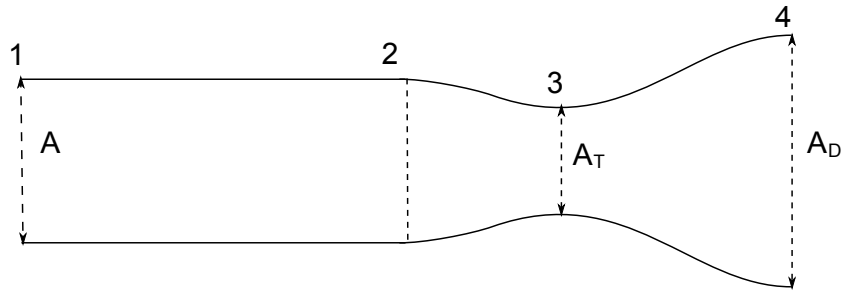


Fig. Q3: Not to scale

(a) For the fully expanded condition $p_4 = p_a$, calculate the required nozzle area ratio A_D/A_T and the exit Mach number M_4. For the same area ratio, calculate the hypothetical ambient pressure p_a for the subsonic isentropic condition when the nozzle is choked at 3.

[20%]

(b) The duct part of the exhaust system 1-2 is converted to an afterburner. If the afterburner is switched on, heat is added uniformly along the duct 1-2, and the air is heated to a stagnation temperature of 2000 K at 2. A_T is made adjustable and due to space constraint, the divergent part 3-4 also needs to be removed. The conditions at inlet are unchanged.

(i) Calculate the required area ratio A_T/A, when the afterburner is switched off.

[20%]

(ii) When the afterburner is switched on, calculate the required A_T/A ratio.

[35%]

(iii) When the afterburner is switched on, calculate the momentum thrust $\dot{m}u_3$ and the pressure thrust $(p_3 - p_a)\,A_T$, both in terms of A. Here $\dot{m}$ is the mass flow rate and u_3 is the exit velocity at 3.

[25%]

END